

CHAPTER ONE: INVENTORIES

1.1. Definition of Inventory

International Accounting Standard 2 *Inventories*

- The following terms are used in this Standard with the meanings specified:
- Inventories are assets:
 - (a) Held for sale in the ordinary course of business
 - (b) In the process of production for such sale; or
 - (c) In the form of materials or supplies to be consumed in the production process or in the rendering of services.

1.2. Inventory Classification

- They are mainly divided into two major categories:
 - ⊙ Inventories of merchandising businesses and
 - ⊙ Inventories of manufacturing businesses
- a) **Inventories of merchandising businesses**
 - ⊙ Are merchandise purchased for resale in the normal course of business.
 - ⊙ These types of inventories are called **merchandise inventories**.
- b) **Inventories of manufacturing business**
 - ⊙ Manufacturing businesses are businesses that produce physical output.
 - ⊙ They normally have three types of inventories. These are:
- **Raw material inventory** – are the basic goods that will be used in production but have not yet been placed into production. **Example**, wood to make a chair or other office furniture's, the steel to make a car etc.
- **Work in process inventory**- is that portion of manufactured inventory that has been placed into the production process but is not yet complete.
- **Finished goods inventory** – manufactured items that are completed and ready for sale.
- **In this unit, only the determination of the inventory of merchandise purchased for resale commonly called merchandise inventory will be discussed.**

1.3 Why is Inventory Control Important?

2. Inventory is a significant asset and for many companies the largest asset.
3. Inventory is central to the main activity of merchandising and manufacturing companies.
4. Mistakes in determining inventory cost can cause critical errors in financial statements.
5. Inventory must be protected from external risks (such as fire and theft) and internal fraud by employees.

1.4 Cost of inventories

- The cost of inventories shall comprise all costs of:
 - ⊙ Purchase
 - ⊙ Costs of conversion and
 - ⊙ Other costs incurred in bringing the inventories to their present location and condition.
- a) **Costs of purchase**
- The costs of purchase of inventories comprise:
 - ⊙ The purchase price

- ⦿ Import duties and other taxes (other than those subsequently recoverable by the entity from the taxing authorities)
- ⦿ Transport
- ⦿ Handling and other costs directly attributable to the acquisition of finished goods, materials and services.
- ⦿ Trade discounts, rebates and other similar items are deducted in determining the costs of purchase.

b) Costs of conversion

- ➔ Costs directly related to the units of production, such as direct labor.

c) Other costs

- ➔ Cost of inventories incurred in bringing the inventories to their present location and condition.
- ➔ **Eg.** Non-production overheads or the costs of designing products for specific customers.

1.5 Inventory System

There are two principal systems of inventory accounting: periodic and perpetual.

Periodic Inventory System	Perpetual inventory system
⦿ There is <u>no continuous</u> record of merchandise inventory ⦿ The revenue from sales is recorded each time a sale is made. ⦿ No entry is made to record the CMS. ⦿ Physical inventory must be taken periodically to determine the CEI and CGS. ⦿ Used by retail enterprises that sell many kinds of low unit cost merchandise such as groceries, drugstores, hardware etc. The journal entries to be prepared are: (1) <u>At the time of purchase of merchandise:</u> Purchases XX Accounts payable or cash XX (2) <u>At the time of sale of merchandise:</u> Accounts receivable or cash XX Sales XX (3) <u>To record purchase returns and allowance:</u> Accounts payable or cash XX Purchase returns and allowance XX (4) <u>To close beginning inventory:</u> Income Summary XX Merchandise inventory XX (5) <u>To record ending inventory:</u> Merchandise inventory XX Income summary XX	⦿ The accounting record <u>continuously disclose</u> the amount of inventory ⦿ <u>Both</u> the sales and CMS should be recorded at the time of each sale. ⦿ No need of physical counting to determine their costs. ⦿ Used by companies that sell items of high unit value, such as appliances or automobiles, tend to use the perpetual inventory system. The journal entries to be prepared are: (1) <u>At the time of purchase of merchandise:</u> Merchandise inventory XX Accounts payable/cash XX (2) <u>At the time of sale of merchandise</u> Accounts receivable or cash XX Sale XX Cost of goods sold XX Merchandise inventory XX (To record the cost of merchandise sold) (3) <u>To record purchase returns and allowances</u> Accounts payable or cash XX Merchandise inventory XX (4) No adjusting entry or closing entry for merchandise inventory is needed at the end of each accounting period under perpetual inventory system.

1.6 Determining Inventory Quantities

- ➔ Determining inventory quantities involves two steps:

- (1) Taking a physical inventory of goods on hand and
- (2) Determining the ownership of goods.

a) Taking a Physical Inventory

- Involves counting, weighing, or measuring each kind of inventory on hand.
 - ◆ When the business is closed or business is slow.
 - ◆ At end of the accounting period.
- Companies take a physical inventory either in perpetual or periodic inventory systems for different reasons:

Perpetual System	Periodic System
⊗ To check the accuracy of inventory records. ⊗ To determine the amount of inventory lost (wasted raw materials, shoplifting, or employee theft).	⊗ To determine the inventory on hand. ⊗ To determine the cost of goods sold for the period.

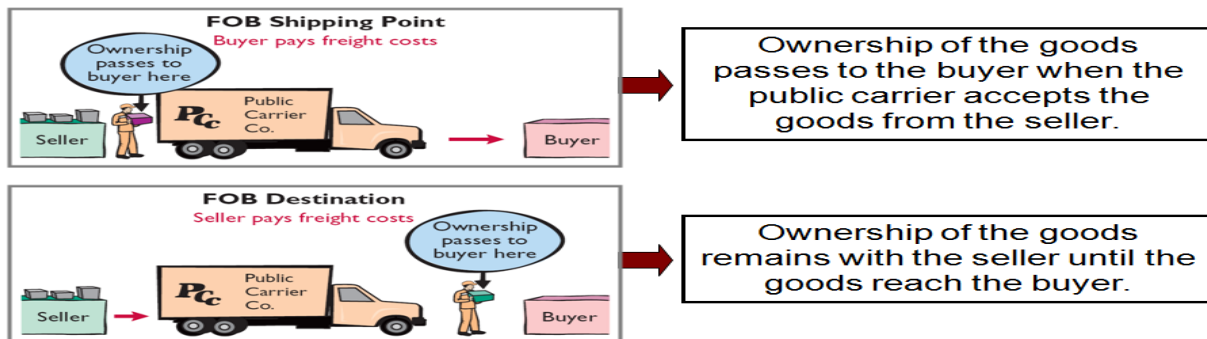
b) Determining Ownership of Goods

1) Goods in Transit

- ◆ Purchased goods not yet received.
- ◆ Sold goods not yet delivered.

Goods in transit should be included in the inventory of the company that has **legal title** to the goods. Legal title is determined by the **terms of sale**.

Goods in Transit



2) Consigned Goods

- In some lines of business, it is common to hold the goods of other parties and try to sell the goods for them for a fee, but without taking ownership of the goods.

Inventory Costing

Unit costs can be applied to quantities on hand using the following costing methods:

- ◆ Specific Identification
 - ◆ First-in, first-out (FIFO)
 - ◆ Average-cost
- } Cost flow Assumption

a) Specific Identification

- **Actual physical flow costing method** in which items still in inventory are specifically costed to arrive at the total cost of the ending inventory.
 - ◆ Practice is relatively rare.

- ◆ Most companies make assumptions (**Cost Flow Assumptions**) about which units were sold.
- ➔ It is inappropriate when there are large numbers of items of inventory that are ordinarily interchangeable.

Illustration: CTZ TV Company purchases three identical 50-inch TVs on different dates at costs of £700, £750, and £800. During the year Crivitz sold two sets at £1,200 each. These facts are summarized below.

Purchases			
February 3	1 TV	at	£700
March 5	1 TV	at	£750
May 22	1 TV	at	£800
Sales			
June 1	2 TVs	for	£2,400 (£1,200 × 2)

- ➔ Assume that if CTZ sold the TVs it purchased on February 3 and May 22 then:
 - ➔ **Compute cost of goods sold and ending inventory?**
- Solution**
- ⦿ Cost of goods sold is £1,500 (£700 + £800) and its ending inventory is £750.
 - ➔ The cost of inventories, other than those dealt with in above, shall be assigned by using the first-in, first-out (FIFO) or weighted average cost formula.
 - ➔ An entity shall use the same cost formula for all inventories having a similar nature and use to the entity.
 - ➔ For inventories with a different nature or use, different cost formulas may be justified.
 - ➔ There are two assumed cost flow methods:
 1. First-in, first-out (FIFO)
 2. Average-cost

1) First-In, First-Out (FIFO)

- ⦿ **Earliest goods** purchased are first to be sold.
- ⦿ Inventory items are sold in the order acquired.
- ⦿ This means the cost flow is in the order in which the expenditures were made.
- ⦿ So, to determine the cost of ending inventory, we have to start from the most recent purchase and continue to the next recent.
- ⦿ Generally good business practice to **sell oldest units first**.

2) Average-Cost

- ⦿ Allocates cost of goods available for sale on the basis of weighted-average unit cost incurred.
- ⦿ Assumes goods are similar in nature.
- ⦿ Applies weighted-average unit cost to the units on hand to determine cost of the ending inventory.

To illustrate the inventory cost flow methods, we will use the data for Houston Electronics' Astro condensers, shown below:

Houston Electronics Astro Condensers				
Date	Explanation	Units	Unit Cost	Total Cost
Jan. 1	Beginning inventory	100	\$10	\$ 1,000
Apr. 15	Purchase	200	11	2,200
Aug. 24	Purchase	300	12	3,600
Nov. 27	Purchase	400	13	5,200
	Total units available for sale	1,000		<u>\$12,000</u>
	Units in ending inventory	450		
	Units sold	<u>550</u>		

- ➔ Compute cost of goods sold and ending inventory through periodic inventory system using

- a) FIFO
b) WA

Solution

- ➔ The cost of goods sold formula in a periodic system is:

(Beginning Inventory + Purchases) - Ending Inventory = Cost of Goods Sold

- a) FIFO

Cost of Goods Available for Sale				
Date	Explanation	Units	Unit Cost	Total Cost
Jan. 1	Beginning inventory	100	\$10	\$ 1,000
Apr. 15	Purchase	200	11	2,200
Aug. 24	Purchase	300	12	3,600
Nov. 27	Purchase	400	13	5,200
	Total	<u>1,000</u>		<u>\$12,000</u>

Step 1: Ending Inventory				Step 2: Cost of Goods Sold	
Date	Units	Unit Cost	Total Cost		
Nov. 27	400	\$13	\$5,200	Cost of goods available for sale	\$12,000
Aug. 24	50	12	600	Less: Ending inventory	<u>5,800</u>
Total	<u>450</u>		<u>\$5,800</u>	Cost of goods sold	<u>\$ 6,200</u>

- b) AVERAGE-COST

Cost of Goods Available for Sale	÷	Total Units Available for Sale	=	Weighted-Average Unit Cost
\$12,000	÷	1,000	=	\$12.00

Cost of Goods Available for Sale				
Date	Explanation	Units	Unit Cost	Total Cost
Jan. 1	Beginning inventory	100	\$10	\$ 1,000
Apr. 15	Purchase	200	11	2,200
Aug. 24	Purchase	300	12	3,600
Nov. 27	Purchase	400	13	5,200
	Total	1,000		<u>\$12,000</u>

Step 1: Ending Inventory			Step 2: Cost of Goods Sold		
\$12,000	÷	1,000	=	\$12.00	Cost of goods available for sale
		Unit		Total	Less: Ending inventory
		Cost		Cost	
450		\$12.00		<u>\$5,400</u>	Cost of goods sold
					<u>\$ 6,600</u>

Inventory Cost Flow Methods in Perpetual Inventory Systems

- What inventory cost flow methods do companies employ if they use a perpetual inventory system?
- Simple—they can use any of the inventory cost flow methods described in the chapter.
- To illustrate the application of the assumed cost flow methods (FIFO and average-cost), we will use the data shown for Houston Electronic's Astro Condenser, shown below:

Houston Electronics Astro Condensers					
Date	Explanation	Units	Units Cost	Total Cost	Balance in Units
1/1	Beginning inventory	100	\$10	\$ 1,000	100
4/15	Purchases	200	11	2,200	300
8/24	Purchases	300	12	3,600	600
9/10	Sale	550			50
11/27	Purchases	400	13	5,200	450
				<u>\$12,000</u>	

- Compute cost of goods sold and ending inventory through perpetual inventory system using
 - a) FIFO } methods
 - b) WA }

Solution

a) First-In, First-Out (FIFO)

Date	Purchases	Cost of Goods Sold	Balance (in units and cost)	
January 1			(100 @ \$10)	\$1,000
April 15	(200 @ \$11) \$2,200		(100 @ \$10)	\$3,200
			(200 @ \$11)	
August 24	(300 @ \$12) \$3,600		(100 @ \$10)	
			(200 @ \$11)	\$6,800
			(300 @ \$12)	
September 10		(100 @ \$10)		\$ 600
		(200 @ \$11)		
		(250 @ \$12)	(50 @ \$12)	
		<u>\$6,200</u>		
November 27	(400 @ \$13) \$5,200		(50 @ \$12)	\$5,800
			(400 @ \$13)	

- ➔ You can see that the results under FIFO in a perpetual system are the **same as in a periodic system**.
- ➔ In both cases, the ending inventory is \$5,800 and cost of goods sold is \$6,200.
- ➔ Regardless of the system, the first costs in are the costs assigned to cost of goods sold.

b) Average-Cost

Date	Purchases	Cost of Goods Sold	Balance (in units and cost)
January 1			(100 @ \$10) \$1,000
April 15	(200 @ \$11) \$2,200		(300 @ \$10.667) \$3,200
August 24	(300 @ \$12) \$3,600		(600 @ \$11.333) \$6,800
September 10		(550 @ \$11.333) \$6,233	(50 @ \$11.333) \$ 567
November 27	(400 @ \$13) \$5,200		(450 @ \$12.816) \$5,767

- ➔ As indicated, Houston Electronics computes a **new average each time it makes a purchase**.

1.7 Lower-of-Cost-or-Net Realizable Value

When the value of inventory is lower than its cost

- ◆ Companies must “write down” the inventory to its net realizable value in the period in which the price decline occurs.
- ◆ Net realizable value refers to the net amount that a company expects to realize (receive) from the sale of inventory (estimated selling price in the normal course of business, less estimated costs to complete and sell).

Illustration: Assume that Gao TV has the following lines of merchandise with costs and net realizable values as indicated.

	Cost	Net Realizable Value	Lower-of-Cost-or-Net Realizable Value
Flat-screen TVs	NT\$60,000	NT\$55,000	NT\$ 55,000
Satellite radios	45,000	52,000	45,000
DVD recorders	48,000	45,000	45,000
DVDs	15,000	14,000	14,000
Total inventory			NT\$159,000

Estimating Inventories

- ➔ We assumed that a company would be able to physically count its inventory.
 - ⦿ What if it cannot?
 - ⦿ What if the inventory were destroyed by fire or flood, for example?
- ➔ In that case, the company would use an estimate.
- ➔ Two circumstances explain why companies sometimes estimate inventories.
 - A casualty such as fire, flood, or earthquake may make it impossible to take a physical inventory.
 - Managers may want monthly or quarterly financial statements, but a physical inventory is taken only annually.
 - The need for estimating inventories occurs primarily with a periodic inventory system because of the absence of perpetual inventory records.

- ➔ There are two widely used methods of estimating inventories:
 - (1) The gross profit method and
 - (2) The retail inventory method.

1) The gross profit method

- ➔ Estimates the cost of ending inventory by applying a gross profit rate to net sales.
 - ⦿ This method is relatively simple, but effective.
 - ⦿ Accountants, auditors, and managers frequently use the gross profit method to test the reasonableness of the ending inventory amount.
 - ⦿ It will detect large errors.
- ➔ To use this method, a company needs to know its:
 - ⦿ Net sales
 - ⦿ Cost of goods available for sale and
 - ⦿ Gross profit rate.

$$\begin{array}{rclcl}
 \text{Step 1:} & \text{Net Sales} & - & \text{Estimated Gross Profit} & = & \text{Estimated Cost of Goods Sold} \\
 & & & \downarrow & & \\
 \text{Step 2:} & \text{Cost of Goods Available for Sale} & - & \text{Estimated Cost of Goods Sold} & = & \text{Estimated Cost of Ending Inventory}
 \end{array}$$

Illustration: K Company's records for January show net sales of \$200,000, beginning inventory \$40,000, and cost of goods purchased \$120,000. The company expects to earn a 30% gross profit rate. Compute the estimated cost of the ending inventory at January 31 under the gross profit method.

Step 1:	
Net sales	\$200,000
Less: Estimated gross profit (30% × \$200,000)	60,000
Estimated cost of goods sold	<u>\$140,000</u>
Step 2:	
Beginning inventory	\$ 40,000
Cost of goods purchased	120,000
Cost of goods available for sale	160,000
Less: Estimated cost of goods sold	140,000
Estimated cost of ending inventory	<u>\$ 20,000</u>

2) Retail Inventory Method

- ➔ Company applies the **cost-to-retail percentage** to ending inventory at retail prices to determine inventory at cost.
- ➔ Under the retail inventory method, a company's records must show both the cost and retail value of the goods available for sale.

Step 1:	Goods Available for Sale at Retail	–	Net Sales	=	Ending Inventory at Retail
Step 2:	Goods Available for Sale at Cost	÷	Goods Available for Sale at Retail	=	Cost-to-Retail Ratio
Step 3:	Ending Inventory at Retail	×	Cost-to-Retail Ratio	=	Estimated Cost of Ending Inventory

Illustration

	At Cost	At Retail
Beginning inventory	\$14,000	\$ 21,500
Goods purchased	61,000	78,500
Goods available for sale	<u>\$75,000</u>	<u>100,000</u>
Net sales		70,000
Step (1) Ending inventory at retail =		<u>\$ 30,000</u>
Step (2) Cost-to-retail ratio \$75,000 ÷ \$100,000 = 75%		
Step (3) Estimated cost of ending inventory = \$30,000 × 75% = \$22,500		<u>\$22,500</u>

1.8 The Effects of Inventories on Financial Statements

- What makes inventories special than any of other accounting items is their multidimensional effect.
- Their effect is not limited in one statement or even within a single period.

Effect of Ending Inventory on Current Period's Financial Statements

- **Ending inventory** is the cost of merchandise on hand at the end of the accounting period. Let us see its effect on current period's financial statements.
- The effect of ending inventory is reflected in both **income statements and balance sheets**.

Income Statement

- a) Ending inventory is used in calculating cost of goods sold in the income statement

Cost of goods (merchandise) sold = Beginning inventory + Net purchase – Ending inventory.

- As you see, ending inventory is a deduction in calculation cost of merchandise sold.
- So, it has an indirect (negative) relationship to cost of merchandise sold, i.e.
- If ending inventory is understated, the cost of merchandise sold will be overstated and if ending inventory is overstated, the cost of merchandise sold will be understated.

- b) The cost of the merchandises sold will then subsequently be used in calculating the gross profit of the enterprise.

Gross Profit = Net sales – Cost of merchandise sold

- Here, the cost of merchandise sold had indirect relationship to gross profit.
- So, the effect of ending inventory on gross profit is the opposites of the effect on cost of merchandise sold.

- That is, if ending inventory is understated, the gross profit will be understated and if ending inventory is overstated, the gross profit will be overstated.
- This is a direct (positive) relationship.

$$\text{Operating income} = \text{Gross Profit} - \text{Expenses}$$

- Gross profit and operating income have direct relationships. Thus, the effect of ending inventory on net income is the same as its effect on gross profit.
- i.e. direct (positive) effect (relationship).

Balance Sheet

- The ending inventory of an organization affects the two major components of a balance sheet:
 - 1) **Current Assets** - Ending inventory is part of current assets, even the largest. Therefore, it has a direct (positive) relationship to current assets. If ending inventory balance is understated (overstated), the total current assets will be understated (overstated). Since current assets are part of total assets, ending inventory has direct relationship to total assets.
 - 2) **Liabilities**: No effect on liabilities. Inventory misstatement has no effect on liabilities.
 - 3) **Owners' equity** - The net income will be transferred to the owners' equity at the end of accounting period. Closing income summary account does this. Therefore, net income has direct relationship with owners' equity at the end of accounting period. The effect-ending inventory on owners' equity is the same as its effect on net income, i.e. if ending inventory is understated (Overstated), the owners' equity will be understated (Overstated).

Illustration:

The first set of statements is based on a correct ending inventory of \$30,000; the second set, on an incorrect ending inventory of \$22,000; and the third set, on an incorrect ending inventory of \$37,000. In all three cases, net sales are \$200,000, merchandise available for sale is \$140,000, and expenses are \$55,000.

Income Statement for the Year		Balance Sheet at End of Year	
1) Inventory at end of period correctly stated at \$30,000			
Net sale	\$200,000	Merchandise inventory	\$30,000
CGS	110,000	Other assets	<u>80,000</u>
Gross profit	\$90,000	Total	<u>\$110,000</u>
Expenses	<u>55,000</u>	Liabilities.....	\$30,000
Net income	<u>\$35,000</u>	Owner's equity	<u>80,000</u>
		Total	<u>\$110,000</u>
2) Inventory at end of period incorrectly stated at \$22,000; (understated by \$8,000).			
Net sales-----	\$200,000	Merchandise inventory	\$22,000
CGS-----	118,000	Other assets	<u>80,000</u>
Gross profit ----	\$82,000	Total.....	<u>\$102,000</u>
Expenses	<u>55,000</u>	Liabilities.....	\$30,000
Net income	<u>\$27,000</u>	Owner's equity ...	<u>72,000</u>
		Total	<u>\$102,000</u>
3) Inventory at end of period incorrectly stated at \$37,000; (overstated by \$7,000).			

Net sales -----	\$200,000	Merchandise inventory	\$37,000
CGS-----	103,000	Other assets ...	<u>80,000</u>
Gross profit -----	\$97,000	Total.....	<u>\$117,000</u>
Expenses ...	<u>55,000</u>	Liabilities.....	\$30,000
Net income	<u>\$42,000</u>	Owner's equity	<u>87,000</u>
		Total	<u>\$117,000</u>

Note that in the illustration the total cost of merchandise available for sale was constant \$140,000. It was the way in which the cost was allocated that varied.

Effect of Ending Inventory on the Following Period's Financial Statement

- The ending inventory of the current period will become the beginning inventory for the following period.
- So, it will have the same effect as beginning inventory of the current period. Let us summarize it.

Income Statement of the Following Period

- a) Ending inventory is used in calculating cost of goods sold in the income statement.

Cost of goods (merchandise) sold = Beginning inventory + Net purchase – Ending inventory.

- b) The cost of the merchandises sold of the following period will then subsequently be used in calculating the gross profit of that period.

Gross Profit = Net sales – Cost of merchandise sold

- The effect of ending inventory on the following period's operating income is the same with that of gross profit.

Balance sheet of the following period

- The ending inventory of the current period will not have an effect on the following period's balance sheet items.
- This is because the balance sheet of the following period reports only ending inventory of that period instead of ending inventory of the previous period.

LIFO INVENTORY METHOD

Last-In-First-Out (LIFO)

- ⊗ **Under IFRS, LIFO is not permitted** for financial reporting purposes.
- ⊗ **Latest goods** purchased are first to be sold.
- ⊗ **Seldom coincides** with actual physical flow of merchandise, except for **goods stored in piles**, such as coal or hay.

Illustration

Cost of Goods Available for Sale				
<u>Date</u>	<u>Explanation</u>	<u>Units</u>	<u>Unit Cost</u>	<u>Total Cost</u>
Jan. 1	Beginning inventory	10	HK\$100	HK\$ 1,000
Apr. 15	Purchase	20	110	2,200
Aug. 24	Purchase	30	120	3,600
Nov. 27	Purchase	40	130	5,200
	Total	<u>100</u>		<u>HK\$12,000</u>
Step 1: Ending Inventory			Step 2: Cost of Goods Sold	
<u>Date</u>	<u>Units</u>	<u>Unit Cost</u>	<u>Total Cost</u>	
Jan. 1	10	HK\$100	HK\$ 1,000	Cost of goods available for sale
Apr. 15	20	110	2,200	Less: Ending inventory
Aug. 24	15	120	1,800	
Total	<u>45</u>		<u>HK\$5,000</u>	Cost of goods sold
				<u>HK\$ 7,000</u>